

Day 15: DAX Time Intelligence Mastery

[DEF] Definition to Remember

Time Intelligence functions in DAX calculate metrics over specific calendar periods including Year-To-Date (YTD), Prior Year comparisons (YoY Growth), and moving rolling averages.

1. Prerequisites for Time Intelligence

- * Dedicated Date Table (Dim_Date) with contiguous, unbroken dates.

2. Year-To-Date (YTD) & Month-To-Date (MTD)

```
// Year To Date Sales
Sales YTD = TOTALYTD([Total Sales], Dim_Date[Date])

// Month To Date Sales
Sales MTD = TOTALMTD([Total Sales], Dim_Date[Date])
```

3. Prior Year & YoY Growth % Calculations

```
// Previous Year Sales (Same Period Last Year)
Sales PY =
CALCULATE(
    [Total Sales],
    SAMEPERIODLASTYEAR(Dim_Date[Date])
)

// Year-over-Year (YoY) Sales Variance
YoY Sales Growth $ = [Total Sales] - [Sales PY]

// Year-over-Year (YoY) Sales Growth %
YoY Sales Growth % =
DIVIDE([YoY Sales Growth $], [Sales PY], 0)
```

4. Moving Rolling Averages (30-Day Rolling Sales)

```
Rolling 30 Days Sales =
CALCULATE(
    [Total Sales],
    DATESINPERIOD(Dim_Date[Date], MAX(Dim_Date[Date]), -30, DAY)
)
```

[MUST-WRITE] Must-Master Skills & Interview Points

1. SAMEPERIODLASTYEAR() vs DATEADD(..., -1, YEAR).
2. How DATESYTD / TOTALYTD automatically handle leap years and fiscal year-ends ("3/31").
3. Formatting YoY growth with dynamic up/down color indicators (/).

[RECALL] Quick Recall

TOTALYTD / TOTALMTD + SAMEPERIODLASTYEAR (Prior Year) + DATESINPERIOD (Rolling Averages)